Intelligent Crop Recommendation System with Explainability, Multi-Season Planning, and Economic Viability Analysis for Sustainable Agriculture

Submitted in partial fulfilment of the requirements

of the term work of the

Mini Project of Semester VI

in

TE INFORMATION TECHNOLGY

(A.Y. 2025-2026)

by

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Primary SDG – 2. Zero Hunger: End hunger, achieve food security, improve nutrition, and promote sustainable agriculture by enabling data-driven crop recommendations for improved productivity and farmer profitability.

Secondary SDG – 12. Responsible Consumption and Production: Ensure sustainable resource management by optimizing fertilizer use, crop rotation, and soil health through intelligent recommendations.

CERTIFICATE

This is to certify that the Mini Project Entitled "Intelligent Crop Recommendation System with Explainability, Multi-Season Planning, and Economic Viability Analysis for Sustainable Agriculture" is bonafied work of

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Submitted to the University of Mumbai in partial fulfilment of the requirement for the submission of the termwork of the Mini Project of TE Information Technology.

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Project Report Approval for Bachelor of Engineering

This project report entitled ***Intelligent Crop Recommendation System with Explainability, Multi-Season Planning, and Economic Viability Analysis for Sustainable Agriculture*** by

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Examiners

Date:

Place:

1. Signature :----------

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Declaration

I declare that this written submission represents my ideas in my own words and where others' ideas or words have been included, I have adequately cited and referenced the original sources. I also declare that I have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in my submission. I understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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1. Abstract:

The agricultural sector faces significant challenges in selecting the most suitable crops due to varying soil conditions, climate changes, and lack of data-driven decision support systems. Traditional crop recommendation approaches often rely on static datasets and do not provide insights into the reasoning behind predictions, economic feasibility, or long-term sustainability. To address these limitations, this project proposes an **Intelligent Crop Recommendation System** that integrates machine learning with explainable and economically viable decision-making.

The system utilizes a **stacking ensemble model** combining Random Forest, XGBoost, and LightGBM to achieve high prediction accuracy based on soil nutrients (N, P, K), temperature, humidity, pH, and rainfall. It incorporates **Explainable AI (SHAP)** to provide transparent reasoning for each recommendation, enhancing user trust. Additionally, the system performs **economic viability analysis** by estimating profit and return on investment using market price data. A **multi-season crop rotation planner** is also included to promote sustainable farming practices and soil health.

The anticipated outcome is a user-friendly web-based application that delivers accurate, explainable, and economically optimized crop recommendations, thereby supporting farmers in making informed and sustainable agricultural decisions.

2. Introduction:

- **Background:** Agriculture remains a vital sector in India, contributing significantly to the economy and employment. However, farmers often face difficulties in selecting the most suitable crops due to changing climatic conditions, soil variability, and lack of access to scientific decision-support systems. With the advancement of **Computer Engineering**, technologies such as **Machine Learning (ML)**, **Data Analytics**, and **Web Applications** are increasingly being applied to solve real-world agricultural problems. Traditional crop recommendation systems are mostly static and do not adapt to real-time environmental conditions or provide personalized suggestions. Moreover, they lack transparency and fail to explain the reasoning behind their predictions. In the current era of **Explainable AI (XAI)** and **data-driven decision-making**, there is a strong need for intelligent systems that not only provide accurate predictions but also justify them. This project aligns with modern trends such as **AI in Agriculture (AgriTech)**, **sustainable farming**, and **smart decision systems**, making it highly relevant in today's technological landscape.

- **Problem Definition:** Existing crop recommendation systems primarily focus on predicting suitable crops based on limited parameters and often act as **black-box models**, offering no explanation for their outputs. Additionally, they do not consider important factors such as **economic profitability**, **market price variations**, and **multi-season crop planning**, which are crucial for farmers' decision-making.

  The key challenges identified are:

  - Lack of **explainability** in predictions (no transparency)
  - Absence of **economic analysis** (profit, ROI not considered)
  - No support for **multi-season crop rotation planning**
  - Use of **static datasets** without real-time environmental integration
  - Limited guidance on **fertilizer usage**

  To overcome these issues, the proposed system aims to develop an **Intelligent Crop Recommendation System** that integrates advanced machine learning techniques with **Explainable AI (SHAP)**, **economic viability analysis**, and **sustainable crop rotation planning**. The system enhances existing solutions by providing accurate, transparent, and practical recommendations, thereby improving decision-making and promoting sustainable agriculture.

3. Objectives:

Primary Objective

- To develop an **Intelligent Crop Recommendation System** that accurately suggests the most suitable crops based on soil nutrients (N, P, K), climatic conditions (temperature, humidity, rainfall), and soil pH using advanced machine learning techniques such as a **stacking ensemble model**.

Secondary Objectives

- **Objective 1:**
  To integrate **Explainable AI (SHAP)** for providing transparent and interpretable explanations for each crop recommendation, thereby increasing user trust and understanding.
- **Objective 2:**
  To incorporate **Economic Viability Analysis** by calculating Return on Investment (ROI), profit estimation, and risk assessment using market price data, enabling financially informed decision-making.
- **Objective 3:**
  To design a **Multi-Season Crop Rotation Planner** that suggests sustainable crop sequences across seasons to maintain soil health and improve long-term agricultural productivity.
- **Objective 4:**
  To implement a **Fertilizer Recommendation System** using KNN to suggest suitable fertilizers along with alternatives based on soil and crop conditions.

- **Objective 5:**
  To develop a **Flask-based Web Application** that provides an interactive and user-friendly interface with real-time weather data integration for practical usability.
- **Objective 6 (Optional):**
  To enhance system accessibility through features like **Progressive Web App (PWA)** support, mobile compatibility, and easy sharing of recommendations.

3. Literature Survey:

S. N.	Title of the Paper	Key Finding	Gap Finding	Research Objective
1	Crop Recommendation using Machine Learning (Ganorkar et al.)	Achieved ~96% accuracy using RF, SVM, Decision Tree	No explainability; model behaves as black box	RO1: Integrate Explainable AI (SHAP) for transparent predictions
2	Crop Recommendation System: A Comparative Study (Acharya et al.)	Ensemble voting improves accuracy up to 98.99%	Uses basic ensemble (voting), not advanced stacking	RO2: Develop Stacking Ensemble Model (RF + XGBoost + LightGBM) for higher performance
3	Precision Agriculture Crop Recommendation (Pudumalar et al.)	Early system using environmental parameters for prediction	Only focuses on prediction; no economic or sustainability factors	RO3: Incorporate Economic Viability Analysis (ROI, profit, risk)
4	ROI-based Crop Recommendation using MLP (Janrao & Shah)	Introduced profitability-based crop selection	No integration with ML prediction and explainability	RO4: Combine ML prediction with economic analysis in one system
5	AI-based Crop Rotation Planning (Fenz	Uses AI for multi-	Rotation planning not integrated	RO5: Develop Multi-Season Crop Rotation

	et al.)	**season crop rotation**	**with recommendation system**	**Planner integrated with ML system**
6	**Crop & Fertilizer Recommendation using ML (Musanase et al.)**	**Provides fertilizer suggestions using ML**	**Not integrated with crop recommendation pipeline**	**RO6: Implement KNN-based Fertilizer Recommendation within unified system**

5. Mapping of Project Outcomes (POs) to Program Outcomes (POs)/Program Specific Outcomes (PSOs

Chapter Guidelines

1. Introduction to the Chapter

The Mini Project serves as a **capstone experience** in the undergraduate engineering curriculum, enabling students to integrate theoretical knowledge and practical skills acquired throughout the program. It provides an opportunity to apply concepts from various domains such as **Machine Learning, Data Analytics, Web Development, and Software Engineering** to solve real-world problems.

This chapter focuses on establishing a formal mapping between the outcomes of the project titled **"Intelligent Crop Recommendation System with Explainability, Multi-Season Planning, and Economic Viability Analysis for Sustainable Agriculture"** and the defined **Program Outcomes (POs)** and **Program Specific Outcomes (PSOs)** as prescribed by accrediting bodies such as the **National Board of Accreditation (NBA), India**.

The mapping is essential to demonstrate how the project contributes to the attainment of key engineering competencies, including problem-solving, design and development of solutions, use of modern tools, sustainability awareness, and ethical responsibility. By aligning the project outcomes with POs and PSOs, this section highlights the relevance of the project in achieving the expected graduate attributes and showcases its role in preparing students for professional engineering practices.

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2. Definition of Program Outcomes (POs) and Program Specific Outcomes (PSOs)

Program Outcomes (POs)

The following are the standard **12 Program Outcomes (POs)** defined for undergraduate engineering programs (as per NBA guidelines):

- **PO1: Engineering Knowledge**
  Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
- **PO2: Problem Analysis**
  Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles.
- **PO3: Design/Development of Solutions**
  Design solutions for complex engineering problems and design system components that meet specified needs with consideration for societal and environmental factors.
- **PO4: Conduct Investigations of Complex Problems**
  Use research-based knowledge and methods including design of experiments, data analysis, and interpretation of data to provide valid conclusions.
- **PO5: Modern Tool Usage**
  Create, select, and apply appropriate techniques, resources, and modern engineering tools such as ML frameworks, programming tools, and visualization tools.
- **PO6: The Engineer and Society**
  Apply reasoning informed by contextual knowledge to assess societal, health, safety, legal, and cultural issues relevant to engineering solutions.
- **PO7: Environment and Sustainability**
  Understand the impact of engineering solutions in societal and environmental contexts and demonstrate knowledge of sustainable development.
- **PO8: Ethics**
  Apply ethical principles and commit to professional ethics and responsibilities in engineering practice.
- **PO9: Individual and Team Work**
  Function effectively as an individual and as a member or leader in diverse teams.
- **PO10: Communication**
  Communicate effectively on complex engineering activities with the engineering community and society through reports, presentations, and documentation.
- **PO11: Project Management and Finance**
  Demonstrate knowledge of engineering and management principles and apply them to manage projects and understand financial aspects.
- **PO12: Life-long Learning**
  Recognize the need for and engage in independent and life-long learning in the context of technological advancements.

Program Specific Outcomes (PSOs)

- PSO1: Intelligent Systems and Data Analytics
  To design, develop, and deploy intelligent systems using AI, machine learning, and data analytics for solving real-world problems across domains. This includes building predictive models, analyzing datasets, automating processes, and leveraging tools such as generative AI, cloud platforms, and edge analytics.
- PSO2: Secure, Connected, Scalable and Sustainable IT Solutions
  To engineer secure, scalable, sustainable, and distributed IT systems using modern technologies such as IoT, 5G, cybersecurity, and cloud-edge infrastructures. This includes developing applications for smart systems, optimizing performance, ensuring security, and deploying robust distributed architectures.

3. Statement of Project Outcomes (Pos)

- **PO_1 (Proj): Design and Develop**
  Design and develop an intelligent crop recommendation system using a **stacking ensemble model (Random Forest, XGBoost, LightGBM)** to accurately predict suitable crops based on soil and climatic parameters.
- **PO_2 (Proj): Implement Explainable AI**
  Implement **SHAP-based Explainable AI techniques** to generate interpretable feature-level explanations for each crop prediction, improving transparency and user trust.
- **PO_3 (Proj): Analyze Economic Viability**
  Analyze and evaluate the **economic feasibility** of recommended crops by computing Return on Investment (ROI), profit estimation, and risk assessment using market price data.
- **PO_4 (Proj): Develop Sustainable Planning Modules**
  Develop and integrate a **multi-season crop rotation planner** and a **KNN-based fertilizer recommendation system** to promote sustainable agriculture and soil health management.
- **PO_5 (Proj): Implement and Deploy Web Application**
  Implement and deploy a **Flask-based web application** with real-time weather data integration, providing an interactive and user-friendly interface for end-users.

4. Mapping Methodology

The correlation between Project Outcomes (PROs) and Program Outcomes (POs) / Program Specific Outcomes (PSOs) is determined using a three-point scale, which is commonly adopted in outcome-based education. This mapping is based on the level of contribution of each project outcome towards achieving the defined program outcomes.

The following scale is used for mapping:

- **3 – High Correlation**: The project outcome directly and significantly addresses the corresponding program outcome.
- **2 – Medium Correlation**: The project outcome partially or indirectly addresses the program outcome.
- **1 – Low Correlation**: The project outcome minimally contributes to the program outcome.
- **'–' (Blank)**: No correlation between the project outcome and the program outcome.

This methodology ensures a systematic evaluation of how the proposed Intelligent Crop Recommendation System with Explainability, Multi-Season Planning, and Economic Viability Analysis contributes to achieving the desired graduate attributes and program-specific competencies.

5. The Mapping Table

Project Outcomes (PROs)

- **PRO1:** Design and develop an intelligent crop recommendation system using machine learning.
- **PRO2:** Implement explainable AI and economic analysis for decision support.
- **PRO3:** Develop a web-based application for real-time crop recommendations.
- **PRO4:** Evaluate system performance and analyze sustainability impact.
- **PRO5:** Demonstrate teamwork, communication, and documentation skills.

PRO–PO–PSO Mapping Table

CO/ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9		PO 10	PO 11	PO 12	PSO1	PSO2
PRO 1	3	3		3	3		1							3	2
PRO 2		3	3	3	3	3	3							3	3
PRO 3				3	3				3		3			2	3
PRO 4	3	3	3	3	3	3	3						3	3	2
PRO 5								3	3		3	3	3		

Mapping Scale

- 3 → High Correlation
- 2 → Medium Correlation
- 1 → Low Correlation
- Blank → No Correlation

This table clearly shows how the project outcomes contribute to achieving program outcomes and program-specific outcomes for the Intelligent Crop Recommendation System.

6. Justification for Key Mappings

The following justifications explain the strong correlations between selected Project Outcomes (PROs) and Program Outcomes (POs), demonstrating how the project contributes to achieving graduate attributes.

PRO1 to PO4 (Design and Development of Solutions): High Correlation (3)
PRO1 involves designing and developing an intelligent crop recommendation system using machine learning techniques. This includes system architecture design, model development, and integration of modules such as explainability and economic analysis. These activities directly address **PO4**, which focuses on requirement analysis, system design, and implementation.

PRO2 to PO5 (Modern Tool Usage): High Correlation (3)
PRO2 focuses on implementing Explainable AI and economic analysis using modern tools such as Python, Scikit-learn, SHAP, XGBoost, and Flask. The use of these advanced tools and technologies for solving real-world agricultural problems strongly aligns with **PO5**, which emphasizes the use of modern engineering tools.

PRO4 to PO7 (Environment and Sustainability): High Correlation (3)
PRO4 evaluates system performance and analyzes sustainable agricultural impact through crop rotation planning, efficient fertilizer usage, and climate-aware recommendations. These features directly contribute to sustainable farming practices and environmental awareness, which strongly aligns with **PO7**.

PRO5 to PO10 (Communication): High Correlation (3)
PRO5 involves preparing documentation, reports, and presentations, as well as demonstrating the system. These activities require effective written and oral communication, directly addressing **PO10**.

These mappings confirm that the project not only addresses technical aspects but also contributes to design, sustainability, tool usage, and communication skills, ensuring alignment with program outcomes.

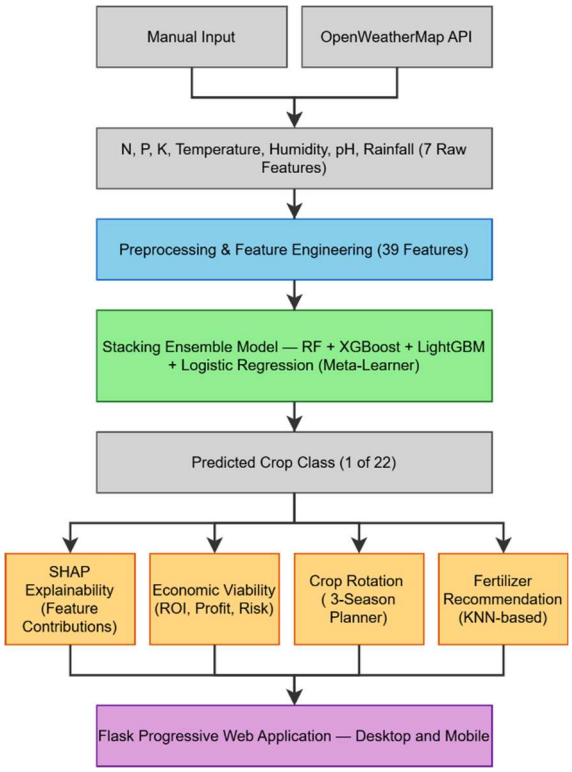
6. Methodology:

6.1. Tools & Technologies:

Category	Tools / Technologies	Purpose / Usage
Programming Languages	Python, HTML5, CSS3, JavaScript	Python for ML & backend, HTML/CSS/JS for frontend development
ML & Data Science Libraries	Pandas, NumPy, Scikit-learn	Data preprocessing, numerical operations, ML model building
	XGBoost, LightGBM	Advanced boosting algorithms for high accuracy prediction
	SHAP	Explainable AI for interpreting model predictions
Data Visualization	Matplotlib, Seaborn	Statistical and graphical data visualization
	Plotly / Chart.js	Interactive charts for web dashboard
Web Framework	Flask, Jinja2	Backend development and dynamic web page rendering
Database & Storage	CSV, Excel, Pickle	Dataset storage and model serialization
External APIs	OpenWeatherMap API	Fetch real-time weather data (temperature, humidity, rainfall)
Development Tools	Jupyter Notebook	Model development and experimentation
	VS Code / PyCharm	Code development environment
	Git & GitHub	Version control and collaboration
Deployment & Additional	Gunicorn	WSGI server for deployment
	Render / Cloud Platform	Hosting the web application
	Progressive Web App (PWA)	Mobile-friendly, installable, offline-capable app

6.2. System Architecture

6.2.1. System Architecture Diagram:



6.2.2. Brief Description: The system takes soil and climatic inputs (N, P, K, temperature, humidity, pH, rainfall) either manually or via real-time weather API, and processes them through a **stacking ensemble model** to predict the most suitable crop. The prediction is further analyzed by major modules including **SHAP-based explainability**, **economic viability analysis**, **crop rotation planner**, and **fertilizer recommendation system**.

The processed results are integrated and displayed through a **Flask-based web application**, ensuring smooth data flow from input → prediction → analysis → user-friendly output.

6.2.3. Key Modules/Blocks:

1. Data Acquisition Module
Collects input parameters such as soil nutrients (N, P, K), temperature, humidity, pH, and rainfall either manually from the user or automatically using the OpenWeatherMap API.

2. Data Preprocessing & Feature Engineering Module
Cleans, normalizes, and transforms input data, and generates additional engineered features to improve model performance and prediction accuracy.

3. Crop Prediction Module (ML Model)
Uses a **stacking ensemble model (Random Forest, XGBoost, LightGBM with meta-learner)** to predict the most suitable crop based on processed input data.

4. Explainability Module (SHAP)
Provides interpretable insights by showing how each feature contributes to the prediction, enhancing transparency and user trust.

5. Economic Analysis Module
Calculates **ROI, profit estimation, and risk level** using market price data to help users choose economically viable crops.

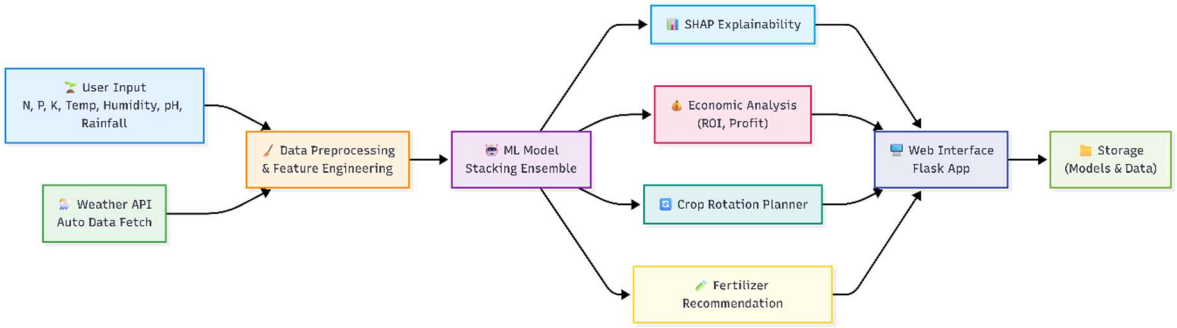
6. Crop Rotation Planning Module
Suggests **multi-season crop sequences** to maintain soil fertility and promote sustainable agriculture practices.

7. Fertilizer Recommendation Module
Uses a **KNN-based approach** to recommend suitable fertilizers along with alternatives based on soil and crop conditions.

8. Web Application Interface (Flask)
Provides a user-friendly interface for input, processing, and visualization of results, ensuring smooth interaction between user and system.

9. Data Storage & Model Management Module
Stores datasets, trained models, and configuration files using CSV and Pickle for efficient retrieval and deployment.

6.2.4. Data Flow:



6.3. Datasets used-

1. Crop Recommendation Dataset (Primary Dataset)

- **Source:** Kaggle – Crop Recommendation Dataset
- **Size:** 2,200 records with 22 crop classes
- **Features:** Nitrogen (N), Phosphorus (P), Potassium (K), Temperature, Humidity, pH, Rainfall

- **Purpose:**
  This is the **core dataset** used to train and test the machine learning models for crop prediction. It provides a balanced dataset with equal samples per crop, ensuring unbiased model performance.
- **Usage in Project:**
  - Used for training multiple ML models (RF, SVM, XGBoost, etc.)
  - Input for feature engineering (39 derived features)
  - Basis for final stacking ensemble model

2. Crop & Fertilizer Dataset

- **Source:** Kaggle
- **Size:** 4,500+ records
- **Features:** Soil parameters, crop type, fertilizer type, environmental conditions
- **Purpose:**
  Used to build the **fertilizer recommendation system** using a KNN-based approach.
- **Usage in Project:**
  - Matches soil and crop conditions to suitable fertilizers
  - Provides alternative fertilizer suggestions
  - Includes practical guidance (e.g., tutorial links)

3. Market Price Dataset

- **Source:** Government of India (Agmarknet / data.gov.in)
- **Features:** Crop name, date, minimum price, maximum price, modal price
- **Purpose:**
  Used for **economic viability analysis**, including profit estimation and ROI calculation.
- **Usage in Project:**
  - Calculates expected profit per acre
  - Determines Return on Investment (ROI)
  - Assesses risk based on price variation

4. Crop Rotation Dataset (Knowledge Base)

- **Source:** Manually created based on agricultural research (ICAR guidelines)
- **Features:** Crop combinations, seasonal compatibility, soil impact
- **Purpose:**
  Supports the **multi-season crop rotation planner**.
- **Usage in Project:**
  - Suggests sustainable crop sequences (e.g., Rice → Wheat → Mung Bean)
  - Maintains soil fertility and nutrient balance
  - Assigns sustainability scores

5. Weather Data (Real-Time API Integration)

- **Source:** OpenWeatherMap API
- **Features:** Temperature, humidity, rainfall

- **Purpose:**
  Provides **real-time environmental data** to enhance prediction accuracy.
- **Usage in Project:**
  - Auto-fills climate parameters
  - Enables location-based recommendations
  - Ensures dynamic and up-to-date inputs

6.4. Implementation Plan:

1. Requirement Analysis & Planning

- Identify system requirements based on agricultural challenges and research gaps.
- Define input parameters (soil nutrients, climate data) and expected outputs (crop, ROI, rotation, fertilizer).
- Finalize tools, technologies, and datasets to be used.

2. Data Collection & Preprocessing

- Collect datasets from sources such as Kaggle and government portals.
- Perform data cleaning (handling missing values, removing duplicates).
- Apply normalization and scaling techniques.
- Perform **feature engineering** to generate additional meaningful features.

3. Model Development

- Train multiple machine learning models: Logistic Regression, Decision Tree, Random Forest, SVM, XGBoost, and LightGBM.
- Evaluate models using metrics such as accuracy, precision, recall, and F1-score.
- Develop a **stacking ensemble model** combining the best-performing models.
- Save trained models using serialization (Pickle).

4. Explainability & Advanced Modules

- Implement **SHAP-based Explainable AI** to interpret model predictions.
- Develop **Economic Analysis module** for ROI and profit estimation.
- Implement **Crop Rotation Planner** for multi-season recommendations.
- Build **Fertilizer Recommendation module** using KNN.

5. Web Application Development

- Develop backend using **Flask** for integrating ML models and APIs.
- Design frontend using HTML, CSS, JavaScript, and Bootstrap.
- Integrate modules to display prediction, explanation, and analysis results.
- Connect with **OpenWeatherMap API** for real-time data input.

6. Testing & Validation

- Perform unit testing for individual modules.
- Conduct integration testing to ensure smooth data flow between modules.
- Validate model predictions using test datasets.
- Evaluate system performance and user interface usability.

7. Deployment

- Deploy the application using platforms like **Render** with **Gunicorn** server.
- Enable **Progressive Web App (PWA)** features for mobile accessibility.
- Ensure proper configuration of APIs and model files.

8. Documentation & Maintenance

- Prepare project documentation including report, diagrams, and user manual.
- Monitor system performance and fix bugs if any.
- Plan future enhancements such as IoT integration and deep learning models.

7. Expected Outcome/ Results in current semester

7.1. Expected Outcome of the Project

The project "Intelligent Crop Recommendation System with Explainability, Multi-Season Planning, and Economic Viability Analysis for Sustainable Agriculture" aims to deliver a comprehensive decision-support system for farmers and stakeholders. The final system will:

- Accurately recommend suitable crops based on soil nutrients and climatic conditions using a stacking ensemble machine learning model.
- Provide transparent explanations (SHAP-based) for each prediction, enabling users to understand the reasoning behind recommendations.
- Evaluate economic feasibility by calculating Return on Investment (ROI), expected profit, and risk levels using real market data.
- Suggest multi-season crop rotation plans to ensure sustainable farming and maintain soil health.
- Recommend appropriate fertilizers using a KNN-based approach with alternative options.
- Offer a user-friendly web application with real-time weather integration for practical usability.

This system addresses key limitations of existing solutions by combining accuracy, explainability, economic analysis, and sustainability into a single integrated platform.

7.2. Results Achieved in the Current Semester

During the current semester, significant progress has been made in the development and validation of the system:

- Successfully collected and preprocessed multiple datasets, including crop, fertilizer, and market price data.
- Developed and evaluated multiple machine learning models, achieving high prediction accuracy (~99.32%) using a stacking ensemble approach.

- Implemented feature engineering techniques, improving model performance with additional derived features.
- Integrated SHAP-based Explainable AI, generating meaningful feature-level explanations for predictions.
- Developed initial versions of economic analysis, crop rotation, and fertilizer recommendation modules.
- Built a functional Flask-based web application prototype capable of taking user input and displaying prediction results.
- Integrated real-time weather data API for dynamic input handling**.**

8. Project Timeline:

- **Week 1–2: Phase 1 – Initial Research & Planning**
  Conduct literature review based on research papers, identify problem statement and research gaps, finalize datasets, tools, and project objectives.
- **Week 3–4: Phase 2 – System Design**
  Design system architecture, define modules (ML model, SHAP, economic analysis, rotation, fertilizer), and prepare data flow diagrams and project structure.
- **Week 5–8: Phase 3 – Development**
  Perform data preprocessing and feature engineering, train multiple ML models, develop stacking ensemble model, implement SHAP explainability, economic analysis, crop rotation, and fertilizer recommendation modules. Build Flask web application and integrate all components.
- **Week 9–10: Phase 4 – Testing & Final Report**
  Conduct unit and integration testing, validate model performance, debug issues, finalize deployment, and prepare project documentation including report, PPT, and demo.

9. Hardware/Software Requirements:

9.1. Hardware Requirements

- **Processor:** Minimum Intel i5 / AMD Ryzen 5 (or equivalent)
- **RAM:** Minimum 8 GB (16 GB recommended for ML model training)
- **Storage:** Minimum 256 GB SSD (for datasets, models, and development tools)
- **System Type:** Laptop/Desktop capable of running Python-based applications
- **Internet Connectivity**: Required for dataset download, API access (weather data), and deployment
- **Optional**: GPU (NVIDIA) for faster model training (not mandatory)

9.2. Software Requirements

- **Operating System:** Windows / Linux / macOS
- **Programming Language:** Python (3.10 or above)
- **IDE / Development Tools:**
  - Jupyter Notebook (for ML development)
  - VS Code / PyCharm (for application development)
- **Libraries & Frameworks:**
  - Pandas, NumPy (data processing)

    - Scikit-learn (machine learning models)
    - XGBoost, LightGBM (advanced ensemble models)
    - SHAP (Explainable AI)
    - Matplotlib, Seaborn, Plotly (data visualization)
- **Web Development:**
    - Flask (backend framework)
    - HTML, CSS, JavaScript, Bootstrap (frontend)
- **APIs:**
    - OpenWeatherMap API (real-time weather data)
- **Version Control:**
    - Git & GitHub
- **Deployment Tools:**
    - Gunicorn (WSGI server)
    - Render / Cloud platform

9. Alignment with Sustainable Development Goals (SDGs)

9.1. Introduction to the Chapter

Engineering solutions play a crucial role in addressing some of the most pressing global challenges such as food security, climate change, resource management, and sustainable development. With the rapid advancement of technology, engineers are increasingly expected to develop innovative systems that not only solve technical problems but also contribute positively to society and the environment.

The United Nations introduced the 17 Sustainable Development Goals (SDGs) as a universal call to action aimed at ending poverty, protecting the planet, and ensuring peace and prosperity for all by the year 2030. These goals provide a global framework for governments, organizations, and individuals to align their efforts toward sustainable progress.

This chapter highlights how the project "Intelligent Crop Recommendation System with Explainability, Multi-Season Planning, and Economic Viability Analysis for Sustainable Agriculture" contributes to these global goals. Although developed as an academic project, the system is designed to address real-world agricultural challenges by promoting sustainable farming practices, economic stability for farmers, and efficient resource utilization.

By mapping the project outcomes to relevant SDGs, this section demonstrates the project's potential to create a meaningful societal and environmental impact, aligning engineering innovation with global sustainability objectives.

9.2. Overview of Relevant Sustainable Development Goals

1. **SDG 2: Zero Hunger**

- **Description**:
  This goal aims to end hunger, achieve food security, improve nutrition, and promote sustainable agriculture by increasing agricultural productivity and supporting farmers.

- **Relevance to Engineering**:
  Engineering plays a vital role in developing smart agricultural systems, improving crop yield, and optimizing resource utilization through technologies like machine learning and data analytics.

- **Project Relevance**:
  The proposed system enhances crop selection accuracy, helping farmers choose the most suitable crops, thereby improving yield and contributing to food security.

2. **SDG 12: Responsible Consumption and Production**

- **Description**:
  This goal focuses on ensuring sustainable consumption and production patterns by efficiently using natural resources and reducing waste.

- **Relevance to Engineering**:
  Engineers design systems that promote efficient resource usage, minimize environmental impact, and support sustainable production processes.

- **Project Relevance**:
  The system promotes efficient use of soil nutrients, water, and fertilizers, and suggests crop rotation strategies, reducing resource wastage and supporting sustainable farming.

3. **SDG 13: Climate Action**

- **Description**:
  This goal aims to take urgent action to combat climate change and its impacts by improving resilience and adaptive capacity.

- **Relevance to Engineering**:
  Engineering solutions help in developing climate-resilient systems, monitoring environmental conditions, and enabling adaptive decision-making.

- **Project Relevance**:
  By integrating real-time weather data and recommending climate-suitable crops, the system helps farmers adapt to changing environmental conditions, thereby supporting climate resilience.

9.3. Primary SDG Alignment

This project primarily contributes to **SDG 2: Zero Hunger**.

The project's focus on developing an **intelligent crop recommendation system using machine learning, explainable AI, and economic analysis** directly supports the targets of this goal, particularly those related to **ensuring sustainable food production systems and improving agricultural productivity**.

By analyzing soil nutrients (N, P, K), climatic conditions (temperature, humidity, rainfall), and soil pH, the system accurately recommends the most suitable crops, thereby helping farmers **maximize yield and reduce the risk of crop failure**. Additionally, the integration of **multi-season crop rotation planning** promotes soil health and long-term agricultural sustainability, while the **economic viability module (ROI and profit analysis)** ensures that farmers make financially beneficial decisions.

Furthermore, the inclusion of **fertilizer recommendations** and **real-time weather data integration** enhances resource efficiency and supports better farm management practices. These features collectively enable farmers to adopt **data-driven and sustainable agricultural practices**, aligning with the core objective of SDG 2, which is to **end hunger, achieve food security, and promote sustainable agriculture**.

9.4. Secondary SDG Alignment

Additionally, this project secondarily supports **SDG 12: Responsible Consumption and Production**. While not the primary focus, the project's use of **data-driven decision-making, fertilizer recommendation, and crop rotation planning** promotes efficient utilization of natural resources such as soil nutrients, water, and fertilizers. By minimizing unnecessary input usage and reducing agricultural waste, the system contributes to sustainable production practices and resource optimization.

Furthermore, the project also supports **SDG 13: Climate Action**. Through the integration of **real-time weather data and climate-aware crop recommendations**, the system enables farmers to adapt to changing environmental conditions. By helping in selecting crops suitable for current and predicted climate conditions, it indirectly enhances resilience to climate variability and supports sustainable agricultural adaptation strategies.

These secondary alignments demonstrate that the project not only addresses food security but also contributes to **environmental sustainability and climate resilience**, reflecting a broader and holistic impact.

9.5. Mapping Table (SDG Contribution Summary)

SDG No.	Sustainable Development Goal	Nature of Contribution	Key Project Feature Enabling Contribution
2	Zero Hunger	Primary	ML-based crop recommendation system improves agricultural productivity and food security through accurate crop selection
12	Responsible Consumption and Production	Secondary	Fertilizer recommendation and crop rotation planning ensure efficient use of resources and reduce wastage
13	Climate Action	Secondary	Integration of real-time weather data and climate-aware crop prediction supports adaptation to changing environmental conditions

9.6. Conclusion on Broader Impact

In conclusion, the project **"Intelligent Crop Recommendation System with Explainability, Multi-Season Planning, and Economic Viability Analysis for Sustainable Agriculture"** demonstrates strong alignment with global sustainability goals. It primarily contributes to **SDG 2: Zero Hunger** by improving agricultural productivity and food security through accurate, data-driven crop recommendations.

Additionally, it supports **SDG 12: Responsible Consumption and Production** and **SDG 13: Climate Action** by promoting efficient resource utilization, sustainable farming practices, and climate-aware decision-making.

If implemented at a larger scale, this system has the potential to significantly enhance **farmer decision-making, reduce crop failures, optimize resource usage, and improve economic outcomes**, especially for small and marginal farmers. It can contribute to building a more **resilient and sustainable agricultural ecosystem**, thereby addressing real-world challenges in food security and environmental sustainability.

This alignment underscores the broader responsibility of engineers to develop solutions that go beyond technical performance. It highlights the importance of creating systems that are **socially impactful, economically viable, and environmentally sustainable**, reinforcing the role of engineering as a key driver in achieving global development goals.

10. Conclusion:

The project **"Intelligent Crop Recommendation System with Explainability, Multi-Season Planning, and Economic Viability Analysis for Sustainable Agriculture"** aims to develop a comprehensive, data-driven decision-support system that integrates **machine learning, explainable AI, economic analysis, and sustainable planning**. The primary goal is to assist farmers in selecting the most suitable crops based on soil and climatic conditions while ensuring transparency, profitability, and long-term sustainability.

From a computer engineering perspective, the project demonstrates the effective application of **advanced machine learning techniques (stacking ensemble), real-time data integration, and web-based deployment** to solve real-world problems. It highlights the role of intelligent systems in transforming traditional domains like agriculture through automation, analytics, and user-centric design.

The anticipated impact of this project lies in promoting **smart agriculture, improving resource efficiency, and enabling informed decision-making**, thereby showcasing how computer engineering solutions can contribute to both technological advancement and societal development.

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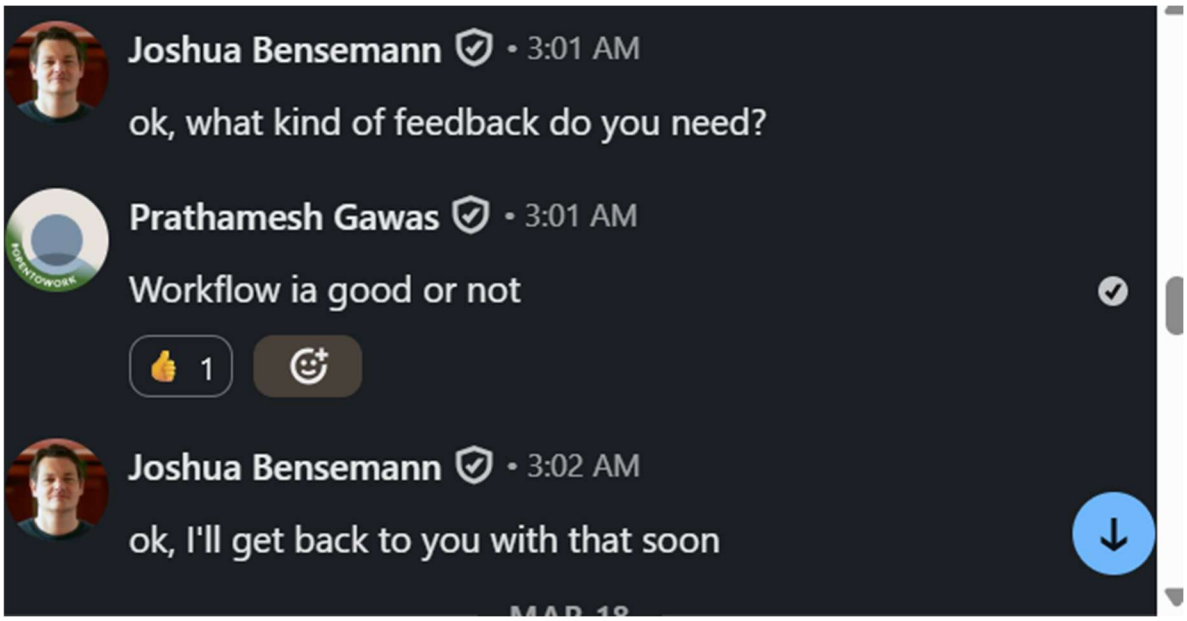
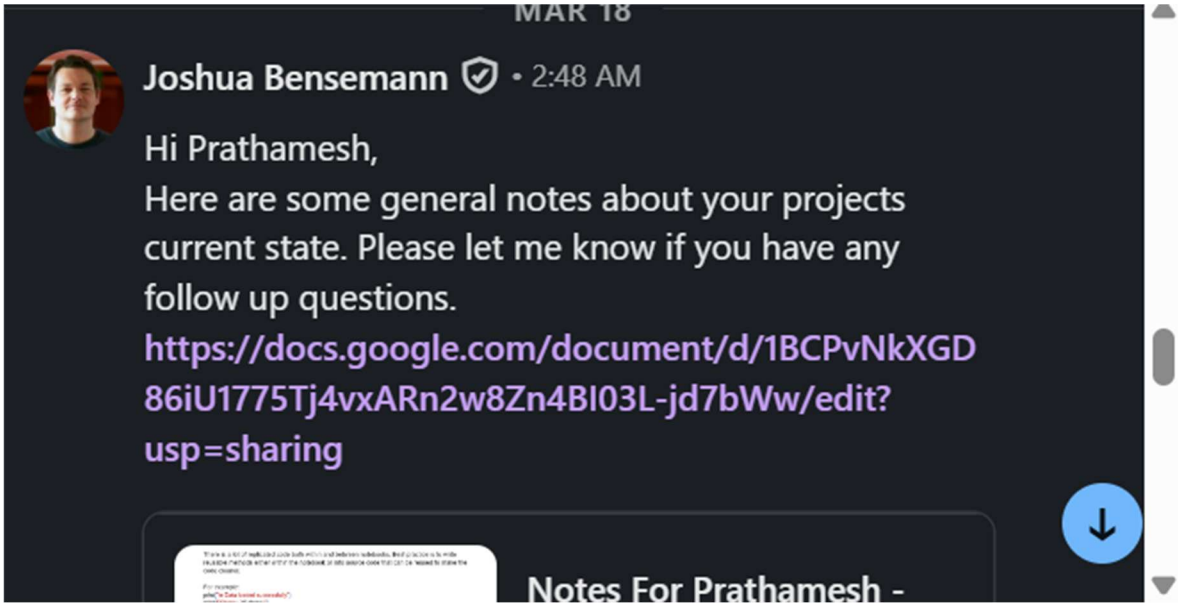
Communication Details – email communication, zoom / google meet screenshots, meeting between Industry mentor and group members along with allocated guide





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Notes for Prathamesh

General Notebook Structure

The notebooks follow a logical order:

- Starts with data checks and exploration
- Next data prep, including feature engineering and train/test splitting
- Then model training, separated by model type
- Finally, a practical analysis of the problem in its intended context

Additional Notes:
Notebook names are prefixed with a number to show intended order, which is very useful.
It would be best to add a 0 to single-digit numbers so 1 becomes 01, this will display the notebooks in order within the folder structure.

Notebook Notes

There is a lot of replicated code both within and between notebooks. Best practice is to write reusable methods either within the notebook or into source code that can be reused to make the code cleaner.

For example:

```
print("\n Data loaded successfully")
print(f"Shape: {df.shape}")
print(f"\nColumns:{list(df.columns)}")
print(f"\nFirst 5 rows:\n{df.head()}")
```

Variations of this appear in notebooks 3, 4, and 5. It would be better to write a load_data method in another file, import it into each notebook and run it.
The sample applies to the Load Prepared Datasets section of all the Model Training Notebooks.
All of this could be turned into reusable code for this project and used in future projects.

Data Science Notes

Be careful about data leakage caused by pre-scaling. For example, in notebook 8, there is a cross-validation score section that performs k-fold cross-validation on the training data. The k-fold is a good idea; however, the training data was scaled beforehand, which could introduce a leak across folds. Instead of scaling during the train-test split notebook, it is best to scale immediately before model training. Between yet, if using sklearn, then the scaling can be used as part of the sklearn.pipeline.Pipeline class, so that it is applied to training and testing directly.